

Derivation of the Cart-Pole Equations of Motion

Jasper Jørgensen

September 2026

1 Nomenclature

Symbol	Meaning	Unit
x	Cart position, measured in the ground frame	m
θ	Pole angle from upright, counter-clockwise positive	rad
x_p, y_p	Position of the pole's centre of mass	m
M	Mass of the cart	kg
m	Mass of the pole	kg
l	Distance from the pivot to the pole's centre of mass	m
I_S	Moment of inertia of the pole about its centre of mass	kg m^2
g	Gravitational acceleration	m s^{-2}
F	Control force applied to the cart	N
dF	External disturbance force at the pole's centre of mass	N
b	Viscous damping coefficient of the cart	N s m^{-1}
H, V	Horizontal and vertical reaction forces at the pivot	N
N	Normal force from the ground on the cart	N

We model the pole as a uniform slender rod. Since l is defined here as *half* the total rod length, its moment of inertia about the centre of mass is given by

$$I_S = \frac{1}{12} m (2l)^2 = \frac{1}{3} m l^2. \quad (1)$$

2 The cart-pole problem: definitions

Looking at the system setup in Figure 1, our first step is to get the coordinates for the pole's center of mass in terms of the cart's position x and the pole angle θ . Since the origin is fixed to the ground (and doesn't move with the cart), x naturally shows up in the equations:

$$x_p = x - l \sin \theta, \quad (2)$$

$$y_p = l \cos \theta. \quad (3)$$

To use Newton's second law, we need accelerations. This means taking the second time derivatives of (2) and (3). Keeping in mind that θ varies with time, applying the chain and

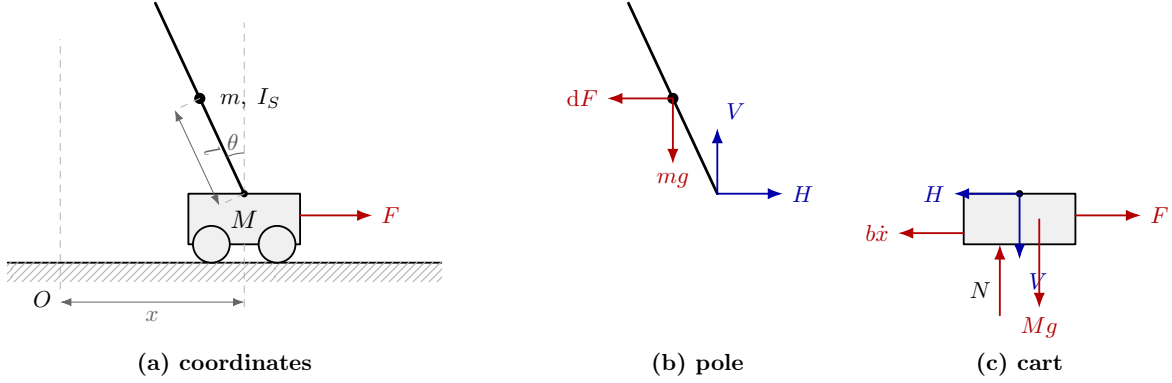


Figure 1: Cart-pole system. (a) Coordinates and conventions, (b) free-body diagram of the pole, (c) free-body diagram of the cart.

product rules yields:

$$\dot{x}_p = \dot{x} - l\dot{\theta} \cos \theta, \quad \ddot{x}_p = \ddot{x} - l\ddot{\theta} \cos \theta + l\dot{\theta}^2 \sin \theta, \quad (4)$$

$$\dot{y}_p = -l\dot{\theta} \sin \theta, \quad \ddot{y}_p = -l\ddot{\theta} \sin \theta - l\dot{\theta}^2 \cos \theta. \quad (5)$$

The $\dot{\theta}^2$ terms and the trig functions, which are exactly what make the cart-pole system nonlinear.

3 The pole

To get the equations of motion, it's easiest to split the system into two free bodies: the pole (Figure 1b) and the cart (Figure 1c). They interact at the pivot through the reaction forces H and V .

Let's write out the Newton–Euler equations for the pole. Starting with the linear momentum balance (the *Schwerpunktsatz*) in the x and y directions:

$$\sum F_x = m \ddot{x}_p \implies H - dF = m \ddot{x}_p, \quad (6)$$

$$\sum F_y = m \ddot{y}_p \implies V - mg = m \ddot{y}_p. \quad (7)$$

Next up is the angular momentum balance (the *Drallsatz*) about the centre of mass. We pick the centre of mass as our reference point on purpose so gravity doesn't create a moment, simplifying things a bit:

$$\sum M_S = I_S \ddot{\theta} \implies I_S \ddot{\theta} = l \sin \theta V + l \cos \theta H. \quad (8)$$

Figuring out the lever arms is just basic trigonometry (see Figure 2). The vector pointing from the centre of mass S down to the pivot P is $(l \sin \theta, -l \cos \theta)$. This means the horizontal force H has a vertical lever arm of $l \cos \theta$, and the vertical force V has a horizontal lever arm of $l \sin \theta$. Since we define θ as positive counter-clockwise, both of these create positive moments.

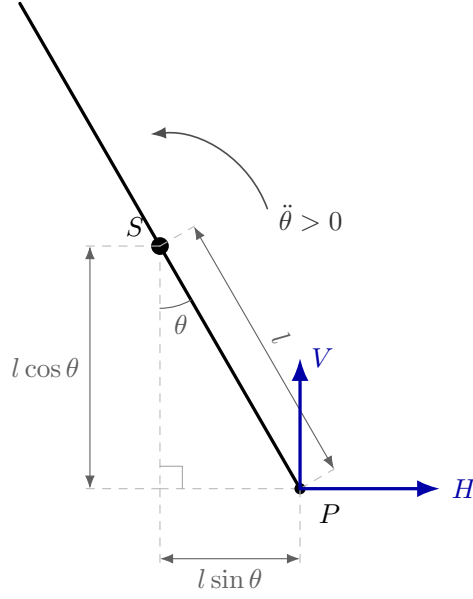


Figure 2: Lever arms of the pivot reactions about the pole's centre of mass.

If we solve (6) and (7) for H and V , and plug in our acceleration terms from (4) and (5), we get:

$$H = dF + m\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta, \quad (9)$$

$$V = mg - ml\ddot{\theta}\sin\theta - ml\dot{\theta}^2\cos\theta. \quad (10)$$

Now we just substitute these expressions for H and V back into our moment equation (8):

$$\begin{aligned} I_S\ddot{\theta} &= mgl\sin\theta - ml^2\ddot{\theta}\sin^2\theta - ml^2\dot{\theta}^2\sin\theta\cos\theta \\ &\quad + dFl\cos\theta + ml\ddot{x}\cos\theta - ml^2\ddot{\theta}\cos^2\theta + ml^2\dot{\theta}^2\sin\theta\cos\theta. \end{aligned}$$

Conveniently, the $\dot{\theta}^2$ terms cancel each other out entirely, and the $\ddot{\theta}$ terms simplify nicely since $\sin^2\theta + \cos^2\theta = 1$. Moving those to the left-hand side leaves us with our first equation of motion:

$$\boxed{(I_S + ml^2)\ddot{\theta} = ml(g\sin\theta + \ddot{x}\cos\theta) + dFl\cos\theta} \quad (11)$$

You'll notice the term $I_S + ml^2$ here. That is simply the pole's moment of inertia about the pivot (thanks to Steiner's parallel axis theorem). It makes physical sense that this shows up, since the pivot is the point the pole is actually rotating around. Using our definition from (1):

$$I_S + ml^2 = \frac{1}{3}ml^2 + ml^2 = \frac{4}{3}ml^2. \quad (12)$$

4 The cart

Since the cart only slides back and forth and doesn't rotate, its angular acceleration is zero, meaning we can neglect the moment balance. Similarly, the vertical force balance just gives

us the normal force N from the ground. We don't actually need either of these to simulate the dynamics.

So, we just look at the horizontal forces on the cart:

$$M\ddot{x} = F - H - b\dot{x}. \quad (13)$$

Plugging in our earlier expression for H (9) and grouping the acceleration terms on the left gives us the second equation of motion:

$$\boxed{(M + m)\ddot{x} - ml \cos \theta \ddot{\theta} = F - b\dot{x} - dF - ml\dot{\theta}^2 \sin \theta} \quad (14)$$

5 Mass matrix form

Right now, equations (11) and (14) are coupled, meaning $\ddot{x}$ and $\ddot{\theta}$ show up in both. To make this useful (e.g., for numerical simulation), we can isolate the acceleration terms on the left and write the whole thing in matrix form:

$$\underbrace{\begin{bmatrix} M + m & -ml \cos \theta \\ -ml \cos \theta & \frac{4}{3}ml^2 \end{bmatrix}}_{\mathbf{A}(\theta)} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} F - b\dot{x} - dF - ml\dot{\theta}^2 \sin \theta \\ mgl \sin \theta + dF l \cos \theta \end{bmatrix}. \quad (15)$$

Here, $\mathbf{A}(\theta)$ acts as the mass matrix of the system. It's essentially the multivariable equivalent of the m in $F = ma$. But because the cart and pole movements are coupled, it's a matrix and importantly, it changes depending on the pole's angle θ .

Those off-diagonal entries ($-ml \cos \theta$) are what actually quantify the coupling between the two bodies. Interestingly, they go to zero when $\theta = \pm\pi/2$. This makes physical sense: if the pole is perfectly horizontal, accelerating the cart doesn't create any moment around the pivot, so the two motions briefly decouple.

The fact that both off-diagonal terms are identical is that mass matrices are always symmetric in classical mechanical systems. Seeing that symmetry pop up in (15) suggests that we haven't messed up the derivation somewhere.

Finally, by inverting this 2×2 matrix, we can solve for $\ddot{x}$ and $\ddot{\theta}$ explicitly. This is exactly what we need to set up the state-space model and numerically integrate the state vector $\mathbf{x} = [x, \dot{x}, \theta, \dot{\theta}]^T$.